PCBC H Parametric & Sequences Test (Practice D) **Name**_____ _____/27

SHOW ALL WORK FOR FULL CREDIT. ***Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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_____ _____

Signature

Solve the following application problem. Report answers to the thousandths. Include units in your answers. (4 points)

7. A soccer ball is kicked from the ground at an angle of 35° with an initial speed of 50 ft/s.

a) Write a set of parametric equations for the motion of the soccer ball. (Recall: $x = (v\cos\theta)t,\ \ y = (v\sin\theta)t - 16t^2 + h$)

b) Determine the maximum height reached by the soccer ball.

c) Determine how long the soccer ball is in the air.

d) Determine how far the soccer ball traveled in the air.

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern. (3 points)

8. $\frac{2}{3}, -\frac{4}{5}, \frac{6}{7}, -\frac{8}{9}, \cdots$

Express the sum using summation notation. (3 points)

9. $-\frac{1}{2}+\frac{1}{4}-\frac{1}{8}+\cdots+\frac{1}{1024}$

Find each sum. Show the properties and formulas you use, then use your calculator. (#10 and #11 are 3 points each, #12 is 4 points)

10. $\sum_{k=1}^{30}(5-2k)$

11. $\sum_{k=1}^{10}\left(k^3-2k^2+1\right)$

12. $\sum_{k=11}^{30} \left(6k^2 + k\right)$

Expand using the Binomial Theorem. (#13 is 3 points, #14 is 4 points)

13. $(x-2)^7$

14. $(2x+3)^5$

Extra practice: more sequence and summation questions

Not part of the 27 points. Worked in the key.

Write out the sum.

E1. $\displaystyle\sum_{k=2}^{n} \frac{(-1)^k}{k(k-1)}$

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern.

E2. $-1,\ \dfrac{3}{4},\ -\dfrac{5}{9},\ \dfrac{7}{16},\ \ldots$

Answer Key

Work on the left, the reason beside it. Boxed = final answer.

Day 2: Calculator

7. Soccer ball: $h = 0$ ft, $\theta = 35°$, $v = 50$ ft/s. Calculator in **degree** mode, Graph in **Parametric**.

a) $x = (50\cos 35°)t$ — *plug v and θ into the given formula; degree symbol required*

$y = (50\sin 35°)t - 16t^2 + 0$ — *+h: starts at 0 ft*

$x_1(t) = 50\cos(35°)\cdot t,\ \ y_1(t) = 50\sin(35°)\cdot t - 16t^2 + 0$ — *what you type in*

$0 \le t \le 2$, tstep 0.01; x : 0 to 90, y : 0 to 16 — *graph doesn't come back down? raise tmax*

b) $\boxed{y \approx 12.851 \text{ ft}}$ — *trace to the top ($t \approx 0.896$); not Analyze, Maximum*

c) $\boxed{t \approx 1.792 \text{ s}}$ — *trace to $y \approx 0$; trace step 0.01, then 0.001*

d) $\boxed{x \approx 73.413 \text{ ft}}$ — *same point, read x*

Units: feet (the speed was ft/s).

8. $\frac{1}{2}, -\frac{2}{5}, \frac{3}{7}, -\frac{4}{9}, \ldots$

top: $2n$, bottom: $2n+1$ — *write 1 2 3 4 above; top and bottom separately*

$+,-,+,- \Rightarrow (-1)^{n+1}$ — *first term positive*

$\boxed{a_n = (-1)^{n+1}\frac{2n}{2n+1}}$ — *check $n = 1$: $\frac{2}{3}$*

9. $-\frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots + \frac{1}{1024}$

bottom 2^k, $-,+,- \Rightarrow (-1)^k$ — *first term negative*

$1024 = 2^{10} \Rightarrow k = 10$

$\boxed{\sum_{k=1}^{10} \frac{(-1)^k}{2^k}}$ — *$\Sigma_{k=1}^{10}\left(-\frac{1}{2}\right)^k$ is the same*

10.

$\sum_{k=1}^{30} 5 - 2\sum_{k=1}^{30} k$ — *(3), (1)*

$5(30) - 2\left[\frac{30(31)}{2}\right]$ — *(5), (6)*

$\boxed{-780}$

11.

$\sum_{k=1}^{10} k^3 - 2\sum_{k=1}^{10} k^2 + \sum_{k=1}^{10} 1$ — *(2), (3), (1)*

$\left[\frac{10(11)}{2}\right]^2 - 2\left[\frac{10(11)(21)}{6}\right] + 10$ — *(8), (7), (5)*

$\boxed{2265}$

12.

$6\sum_{k=11}^{30} k^2 + \sum_{k=11}^{30} k$ — *(2), (1)*

$6\left[\sum_{k=1}^{30} k^2 - \sum_{k=1}^{10} k^2\right] + \left[\sum_{k=1}^{30} k - \sum_{k=1}^{10} k\right]$ — *(4) on each: starts at 11, so $j = 10$*

$6\left[\frac{30(31)(61)}{6} - \frac{10(11)(21)}{6}\right] + \left[\frac{30(31)}{2} - \frac{10(11)}{2}\right]$ — *(7), (6)*

$\boxed{54830}$

13. $(x-2)^7$: $n = 7$, row 1 7 21 35 35 21 7 1. The sign goes with its term.

$1x^7(-2)^0 + 7x^6(-2)^1 + 21x^5(-2)^2 + 35x^4(-2)^3$ — *blanks first: first term's powers count down, second term's count up*

$+35x^3(-2)^4 + 21x^2(-2)^5 + 7x^1(-2)^6 + 1x^0(-2)^7$

$\boxed{x^7 - 14x^6 + 84x^5 - 280x^4 + 560x^3 - 672x^2 + 448x - 128}$ — *second term negative: odd powers of -2 are negative, so signs alternate*

14. $(2x+3)^5$: $n = 5$, row 1 5 10 10 5 1. The sign goes with its term.

$1(2x)^5(3)^0 + 5(2x)^4(3)^1 + 10(2x)^3(3)^2 + 10(2x)^2(3)^3$ — *blanks first: first term's powers count down, second term's count up*

$+5(2x)^1(3)^4 + 1(2x)^0(3)^5$

$(2x)^5 = 32x^5, \ldots$ — *the whole first term gets the power*

$\boxed{32x^5 + 240x^4 + 720x^3 + 1080x^2 + 810x + 243}$ — *second term positive: every term positive*

Extra practice (not scored)

E1. $\sum_{k=2}^{n} \frac{(-1)^k}{k(k-1)}$

$\frac{(-1)^2}{2(1)} + \frac{(-1)^3}{3(2)} + \frac{(-1)^4}{4(3)} + \cdots + \frac{(-1)^n}{n(n-1)}$ — *start at $k = 2$*

$\boxed{\frac{1}{2} - \frac{1}{6} + \frac{1}{12} - \cdots + \frac{(-1)^n}{n(n-1)}}$ — *don't add*

E2. $-\frac{1}{1}, \frac{2}{4}, -\frac{3}{9}, \frac{4}{16}, \ldots$

top: $2n-1$, bottom: n^2 — *write 1 2 3 4 above; top and bottom separately*

$-, +, -, + \Rightarrow (-1)^n$ — *first term negative*

$\boxed{a_n = (-1)^n \frac{2n-1}{n^2}}$ — *write -1 as $-\frac{1}{1}$: odd tops, square bottoms*